Fertility Inertia under Ambiguity Aversion

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Abstract

High family-policy spending often produces only small changes in fertility. I study this problem in a Barro–Becker model. Parents value descendants' utility, dislike risk in its total value, and assess family plans by their worst-case value. They are uncertain about how well public news predicts children's outcomes. Information inertia creates an interval of bad news in which welfare falls but locally optimal fertility, investment per child, and consumption remain unchanged. Reliable subsidies can still raise fertility. An explicit policy solution discounts uncertain benefits by the lowest possible delivery rate. Thus, unclear delivery requires larger promises for the same fertility response. More effective support raises fertility and shifts and widens the inertia interval. Less uncertainty about signal informativeness narrows it for a fixed family plan. This suggests two policy dimensions: generosity and legibility. The exact inertia result is limited to maxmin preferences, continuous fertility, and a local normal representation.

Keywords: Fertility; family policy; legibility; ambiguity aversion; information quality

JEL classification: J13, J18, D81

1 Introduction

Parents can become more worried about their children's future and keep the same family plan. The same parents can choose a larger family when they receive more reliable financial support. This paper explains how both responses can occur. The explanation uses uncertainty about the meaning of public news for one family's future. Some bad news lowers welfare but leaves fertility, investment per child, and consumption unchanged. I call this result *fertility inertia*. A reliable child benefit can still raise fertility within the inertia interval.

Experience with family policy motivates this question. South Korea's total fertility rate fell from 1.13 in 2006 to 0.72 in 2023. During this period, the government expanded policies to support births (Statistics Korea, 2024).[1] This record does not establish that policy failed. Fertility could have fallen

[1]The reported spending total in early 2024 was approximately 380 trillion won. It covered the government's Basic Plans

further without these policies. Studies find fertility responses to birth bonuses, child subsidies, parental benefits, and childcare expansion (Milligan, 2005; González, 2013; Cohen et al., 2013; Bauernschuster et al., 2016; Raute, 2019; Kim, 2024). These studies do not give a common comparison of births per dollar. They also do not identify the separate effect of making benefits easier to verify. Research examines uncertainty, accounts of the future, and trust in institutions (Vignoli et al., 2020; Guetto et al., 2022; Kwag et al., 2024). These findings support a model that includes parents' own expectations.

The household chooses consumption, fertility, and investment in each child. Following Becker and Barro (1988) and Barro and Becker (1989), parents value their children's utility. Their children then value the next generation's utility. This link forms the dynasty. I add two assumptions about uncertainty. First, parents dislike risk in their descendants' total utility. Children share economic and policy conditions. A larger family therefore puts more dynastic utility at risk. Second, parents do not know how well public news predicts their children's future. They consider a range of signal informativeness. For each plan, they select the value that makes that plan least valuable. This is the *maxmin rule*: choose the plan with the highest worst-case value. Each plan has its own worst-case belief.

The analysis has three steps. Section 2 defines the recursive model. Each generation solves the same type of problem. The section establishes a bounded value function and derives the optimality conditions. Section 3 assumes a local normal distribution for the next generation's value. It shows when worse news leaves fertility and investment unchanged. A special case then fixes investment and uses linear consumption utility and a linear fertility weight. This case gives an explicit fertility rule and policy formulas. The normal distribution is an additional assumption. The recursive existence theorem does not establish it. Exact inertia uses this normal assumption. The explicit policy formulas also require fixed investment and linear preferences.

Bad news can reduce expected outcomes and reduce uncertainty about those outcomes. Consider bad news about future labor-market conditions. If the signal is highly informative, parents expect worse outcomes for their children. Those outcomes also become more predictable. If the signal has low informativeness, the expected loss is smaller. However, more risk remains. The parent compares these two costs. The worst-case belief depends on how much dynastic utility the plan puts at risk. For moderately bad news, the worst-case informativeness can lie strictly between its bounds. It can then change with the signal and the proposed plan.

The first main result concerns joint fertility and investment choices. The parent compares her preferred plan with nearby plans. A slightly smaller family leaves more resources for current consumption. More investment per child can improve future outcomes at a cost today. Worse news about her children's future jobs lowers the value of a family plan. A change in her choice requires a change in the ranking of plans.

Suppose plan $\mathcal{A}$ has a larger family than nearby plan $\mathcal{B}$. Both plans have the same investment per

on Low Fertility and Aged Society (Al Jazeera, 2024). This total includes many types of policy. It does not measure the transfers that prospective parents received.

child. Initially, $\mathcal{A}$ has higher value. Plan $\mathcal{B}$ saves resources today but gives up some dynastic utility. The parent evaluates each plan under its own worst-case belief. These beliefs can differ. Plan $\mathcal{A}$ puts more utility at risk. A belief that leaves more common risk can therefore be especially costly. Plan $\mathcal{B}$ puts less utility at risk. Its worst-case belief can assign more weight to the expected loss from the signal.

In the local normal model, worse news can subtract the same value from both plans. Each plan must first be evaluated under its own worst-case belief. For example, suppose $\mathcal{A}$ has value 10 and $\mathcal{B}$ has value 9. These values include consumption utility and the risk-adjusted value of descendants. A loss of 2 from each plan leaves values of 8 and 7. Welfare falls under both plans. Plan $\mathcal{A}$ still has one more unit of value. A change to $\mathcal{B}$ cannot recover the common loss.

Proposition 4 applies this comparison to every nearby change in fertility and investment. Bad news lowers welfare but preserves the ranking of nearby plans. Optimal fertility, investment, and consumption therefore stay locally unchanged. The result requires interior worst-case beliefs for the plans in the comparison. It also holds current income and child costs fixed. If news reduces current income, the budget can force the household to change its plan.

The explicit example shows the response outside the inertia interval. Parents evaluate good news at the lowest allowed informativeness. They evaluate very bad news at the highest allowed informativeness. Between these cases, the worst-case belief adjusts and fertility stays unchanged. For an interior fertility choice, the policy has three parts. Fertility responds strongly to very bad news. It stays constant in the inertia interval. It responds less strongly to good news (Corollary 2). For a fixed plan, a wider informativeness range widens the inertia interval.

The second main result concerns reliable financial support. A reliable subsidy lowers the cost of a child. It can raise fertility even when the household does not respond to news. The response depends on the payment that parents can rely on receiving. For example, parents may not know whether they qualify for a childcare subsidy or whether a place is available. They must choose their family plan before they know these outcomes. I use *legibility* to mean how well parents can determine the support available when they choose their plan.

Parents use the lowest payment they consider possible. Suppose the government promises 100 units per child. Parents consider delivery of at least 50 certain, but the remaining amount is uncertain. They choose the same fertility as under a certain payment of 50. The government can later pay 100. However, parents did not rely on the extra 50 when they chose their plan.

Proposition 5 establishes this equivalence for the policy model. If the lowest delivery rate is one half, a promise of 200 has the same effect as a certain payment of 100. A larger promise raises fertility only when it raises reliable support.

Public news and financial support affect different parts of the problem. News gives information about future conditions, such as children's job prospects. It leaves current income and child costs fixed.

Inside the inertia interval, worse news subtracts the same value from nearby plans. Their ranking stays unchanged. A reliable payment per child changes the budget. A plan with more children receives more total support. The payment therefore changes the relative cost of plans and can change their ranking. Under Proposition 5, a small increase in the benefit raises fertility inside the inertia interval if the lowest delivery rate is positive.

More reliable support raises fertility and common-risk exposure. In the explicit model, it also shifts and widens the inertia interval. A narrower informativeness range reduces its width for a fixed plan. A reform that improves delivery and information narrows the interval only if the information effect exceeds the exposure effect. The width alone does not determine the probability of inertia. It also does not determine the response at every signal.

The exact inertia result uses maxmin preferences, continuous fertility, and a local normal representation. The paper does not establish the result under smooth ambiguity or integer fertility. These alternatives remain topics for future work.

Related literature Illeditsch et al. (2021) show how risk and ambiguity can make portfolio choices insensitive to news. I apply this information-inertia mechanism to fertility. Family size determines total exposure to common risk. Investment changes expected value and risk. The analysis covers joint fertility and investment choices. It also shows why a reliable transfer can change choices that do not respond to news. Related work studies unequal responses to good and bad news in financial markets (Epstein and Schneider, 2008; Caskey, 2009; Ju and Miao, 2012). The formal methods use maxmin utility (Gilboa and Schmeidler, 1989), recursive ambiguity (Epstein and Schneider, 2003), risk-sensitive and robust control (Hansen and Sargent, 1995, 2001, 2008).

The fertility model builds on the quantity–quality tradeoff (Becker and Lewis, 1973) and dynamic fertility choice (Razin and Ben-Zion, 1975). Related work connects fertility with growth, human capital, and inequality (Galor and Weil, 2000; de la Croix and Doepke, 2003; Doepke and Zilibotti, 2005; Manuelli and Seshadri, 2009). Other studies examine income, employment, and other fertility risks (Ranjan, 1999; Sommer, 2016; Doepke et al., 2023). I retain the Barro–Becker link between generations and add risk-sensitive preferences over total descendant utility. Inattention, adjustment costs, and social norms can also slow changes in fertility. In this model, parents process the news and can still keep the same family plan.

Guide to the analysis Section 2 defines preferences and the recursive structure. Section 3 first uses known informativeness to separate the mean effect from the reduction in risk. It then allows the worst-case belief to change and derives Proposition 4. Section 3.5 gives the explicit example in Figure 1. Section 3.6 gives the policy results. Section 3.7 states the model's limitations. Section 4 concludes. Appendix A gives the proofs and supporting calculations.

2 Dynastic Fertility with Risk Sensitivity and Ambiguous Signals

Parents choose current consumption, fertility, and investment per child. Their children face the same choices in the next generation. This section adds risk sensitivity and uncertain signal informativeness to the Barro–Becker model. I define the household problem, establish a recursive solution, and derive optimality conditions. Section 3 uses an additional normal assumption to study inertia.

2.1 Environment

The state. Time is discrete, and each period corresponds to one generation. A parent enters the period with state

$$(x_t, s_t) \in \mathcal{X} \times \mathcal{S}.$$

The vector x_t records household and aggregate conditions. It includes assets, human capital, age, economic conditions, and policy variables. The scalar s_t is public news about the next generation's environment. It can concern housing, childcare, jobs, or family policy. A larger s_t means better news. Current resources are $m_t \equiv y_t + a_t$. Income y_t and assets a_t depend on x_t.

The controls. The parent chooses consumption $c_t \geq 0$, fertility $n_t \geq 0$, and investment or a transfer $z_t \geq 0$ per child. Fertility is the quantity choice. Investment per child is the quality choice. I allow continuous fertility. A fractional value can represent the intensity or timing of an additional birth plan. The household budget is

$$c_t + n_t(q_t + z_t) \leq m_t, \tag{1}$$

where $q_t > 0$ is the direct cost per child. Total spending per child is $q_t + z_t$.

The transition and the ambiguous parameter. The next generation's state is generated by

$$(x_{t+1}, s_{t+1}) = G(x_t, s_t, z_t, \varepsilon_{t+1}; \rho). \tag{2}$$

The parent chooses her plan before the shock ε_{t+1} occurs. The shock distribution does not depend on her choices. The parameter ρ identifies a probability model for future outcomes. It measures how well current news predicts the next generation's continuation value. Continuation value is utility from that point onward. The parent knows only an interval for signal informativeness:

$$\rho \in [\underline{\rho}, \overline{\rho}], \qquad 0 < \underline{\rho} < \overline{\rho} < 1, \tag{3}$$

The parent assigns no probabilities to values in this interval. For example, national childcare news can affect parents as a group. It may give little information about which family will receive a place. Uncertainty about the probability model is *Knightian ambiguity*. Risk over ε_{t+1} occurs within a specified model. I write expectations under model ρ and investment z_t as $\mathbb{E}_{\rho,z_t}[\,\cdot \mid x_t, s_t]$. I omit the subscripts when their meaning is clear.

Timing. The parent observes s_t. She then chooses (c_t, n_t, z_t). Finally, she observes ε_{t+1}. She chooses before she knows the shock and while signal informativeness remains uncertain.

The dynastic objective. Let

$$V' \equiv V(x_{t+1}, s_{t+1})$$

denote a child's lifetime continuation value. This value includes the child's future fertility choices. Each generation uses the same value function. The model therefore determines this value. The parent assigns it weight $\Phi(n_t)$, following Barro–Becker, where

$$\Phi(0) = 0, \qquad \Phi'(n) > 0, \qquad \Phi''(n) \leq 0.$$

A childless plan has no dynastic continuation payoff. More children increase the weight on future generations. The marginal increase in that weight does not rise with fertility. A common specification is $\Phi(n) = n^{1-\eta}/(1-\eta)$ for $\eta \in [0, 1)$. At $\eta = 0$, it gives $\Phi(n) = n$. The analysis allows other functions with these properties.

2.2 Risk-sensitive continuation value

Relation to Barro–Becker altruism. In Becker and Barro (1988) and Barro and Becker (1989), parents value current consumption and their children's utility. With uncertainty, an expected-utility continuation term is $\beta\Phi(n_t)\mathbb{E}[V' \mid x_t, s_t]$. The parent averages possible continuation values. Concave consumption utility can already make consumption risk costly. I add risk sensitivity over total descendant utility. This changes the evaluation of uncertain outcomes. Under certainty, the link between generations stays unchanged.

At the time of choice, V' is uncertain. I use an exponential *certainty equivalent* to evaluate this risk. A certainty equivalent is a certain payoff with the same value as the uncertain payoff. For any future payoff Y, define

$$\mathcal{C}(Y \mid x_t, s_t) = \mathcal{C}_{\gamma,\rho,z_t}(Y \mid x_t, s_t) \equiv -\frac{1}{\gamma} \log \mathbb{E}\left[\exp(-\gamma Y) \mid x_t, s_t\right], \qquad \gamma > 0. \tag{4}$$

This formula uses constant absolute risk aversion (CARA). The parameter γ measures risk aversion.

A larger γ increases the expected value the parent will give up to avoid risk. As $\gamma \to 0$, the certainty equivalent approaches the expectation. The formula is standard in risk-sensitive control (Hansen and Sargent, 1995; Tallarini, 2000). For a normal payoff, it equals the mean less a variance penalty.

Lemma 1 (CARA certainty equivalent)**.** *Consider a CARA agent with utility* $U(Y) = -\exp(-\gamma Y)$. *Under model* ρ, $\mathcal{C}_{\gamma,\rho,z_t}(Y \mid x_t, s_t)$ *is the certain payoff equivalent to* Y. *If* $Y \mid x_t, s_t, \rho \sim N(\mu_Y, \sigma_Y^2)$, *the certainty equivalent has the exact linear–quadratic form*

$$\mathcal{C}(Y \mid x_t, s_t) = \mu_Y - \frac{\gamma}{2}\sigma_Y^2.$$

For general Y, *define the cumulant-generating function* $\kappa(t) = \log \mathbb{E}e^{tY}$. *Suppose it is finite and twice differentiable near* $t = 0$. *A small-*γ *expansion gives* $\mathcal{C}(Y) = \mathbb{E}[Y] - \frac{\gamma}{2}\operatorname{Var}(Y) + o(\gamma)$. *Consequently,* $\lim_{\gamma \to 0} \mathcal{C}(Y) = \mathbb{E}[Y]$.

Appendix A.1 derives this result. For a normal payoff, the risk penalty equals $\gamma/2$ times the variance. For a general payoff, this formula gives the leading approximation when γ is small.

Parents evaluate the family's total risk. The parent evaluates total descendant utility in one certainty equivalent. Evaluating each child's value first and then scaling it can give a different result. For $\Phi(n) > 0$,

$$\mathcal{C}_\gamma[\Phi(n)V'] = \Phi(n)\mathcal{C}_{\gamma\Phi(n)}[V'] \neq \Phi(n)\mathcal{C}_\gamma[V'] \quad \text{in general.}$$

Under conditional normality, the model's aggregate evaluation is

$$\mathcal{C}_\gamma[\Phi(n)V'] = \Phi(n)\mathbb{E}[V'] - \frac{\gamma}{2}\Phi(n)^2 \operatorname{Var}(V'),$$

Evaluating a child's value first and then scaling it gives a risk penalty proportional to $\Phi(n)$. Evaluating total descendant utility gives a penalty proportional to $\Phi(n)^2$. Children share a common shock. More children therefore increase exposure to the same future risk. This assumption makes worst-case informativeness depend on family size. The mean–variance formula is exact for exponential preferences and a normal payoff. It is an additional specification. It does not approximate the original Barro–Becker objective.

For comparison, consider expected continuation utility in the normal model: $\Phi(n)[\mu(z) + \sigma(z)\rho s]$. For a positive family size, the worst case uses the lowest ρ when $s > 0$. It uses the highest ρ when $s < 0$. There is no variance penalty to oppose the mean effect. The worst case therefore selects a bound. This comparison applies to the assumed normal model. Other models with nonlinear continuation values can have other sources of inertia.

The recursive problem. The payoff to fertility is the scaled continuation value

$$Y_{t+1}(n_t, z_t) = \Phi(n_t)V(x_{t+1}, s_{t+1}). \tag{5}$$

The parent computes each plan's certainty equivalent under every admissible model. She selects the lowest value for that plan. This is the maxmin rule of Gilboa and Schmeidler (1989). She then adds consumption utility and chooses the best plan. The recursive problem is

$$V(x_t, s_t) = \max_{c_t, n_t, z_t} \left\{ u(c_t) + \beta \min_{\rho \in [\underline{\rho}, \overline{\rho}]} \mathcal{C}_{\gamma,\rho,z_t} \left[\Phi(n_t)V(x_{t+1}, s_{t+1}) \mid x_t, s_t \right] \right\} \tag{6}$$

subject to (1). Current utility satisfies $u' > 0$ and $u'' < 0$. The factor $\beta \in (0, 1)$ discounts the next generation's value. The child's continuation value uses the same value function one generation later. This makes the equation recursive. Risk sensitivity and ambiguity affect this value function. Under certainty, the continuation term is $\beta\Phi(n_t)V'$.

The parent selects a worst-case ρ for each proposed (n_t, z_t). I call this minimization a choice by *nature*. Nature is a mathematical method for selecting the worst model. It does not represent another household or the government.

Economic explanation. The certainty equivalent measures risk costs within a model. The minimum over ρ accounts for uncertainty about which model applies. Both evaluations depend on the plan. A larger family puts more dynastic utility at risk. For moderately bad news, a low-informativeness model can be worst because it leaves more risk. With fewer children, the expected loss can matter more relative to risk. A high-informativeness model can then be worse. These adjustments can leave the chosen plan unchanged. Section 3 states the conditions.

2.3 Recursive problem and solution concept

The value function appears on both sides of (6). A solution must make these sides equal. Define the Bellman operator T by

$$(TV)(x, s) = \max_{c,n,z} \left\{ u(c) + \beta \min_{\rho \in [\underline{\rho}, \overline{\rho}]} \mathcal{C}_{\gamma,\rho,z} \left[\Phi(n)V(x', s') \mid x, s \right] \right\} \tag{7}$$

subject to

$$c + n\big(q(x, s) + z\big) \leq m(x, s).$$

A solution is a fixed point $V = TV$ together with policy functions

$$c^*(x,s), \qquad n^*(x,s), \qquad z^*(x,s),$$

and a worst-case belief policy

$$\rho^*(x,s) \in [\underline{\rho}, \overline{\rho}].$$

Assumption 1 (Regularity)**.** *(i) $\mathcal{X} \times \mathcal{S}$ is a Polish space.*

(ii) Investment per child is bounded, $z \in [0, \bar{z}]$. The feasible correspondence is $\Gamma(x,s) = \{(c,n,z) \geq 0 : c + n(q(x,s) + z) \leq m(x,s),\ z \leq \bar{z}\}$. It is nonempty, compact-valued, and continuous.

(iii) u is continuous, increasing, concave, and bounded on the graph of Γ.

Specifically, $\sup_{(x,s),\, a \in \Gamma(x,s)} |u(c)| < \infty$.

(iv) Φ is continuous and increasing. It satisfies $\overline{\Phi} \equiv \sup_{n \in \Gamma_n} \Phi(n) < \infty$. Here Γ_n is the compact set of feasible fertility levels. Also, $\beta\overline{\Phi} < 1$.

(v) The transition law $P_{\rho,z}(\cdot \mid x,s)$ is weakly continuous (Feller) in (x,s,z) and continuous in ρ. The innovation law for ε_{t+1} does not depend on the controls. The subscript z in $\mathbb{E}_{\rho,z}$ thus records only the effect of z on the successor state through G.

The operator T acts on bounded continuous functions on $\mathcal{X} \times \mathcal{S}$. These functions form a Banach space under the supremum norm.

These conditions support the existence proof. Dynamic consistency is a separate issue. A plan is dynamically consistent if later evaluation does not change it solely because the evaluation date changes. Repeated worst-case calculations can conflict if ρ is fixed for all time. Epstein and Schneider (2003) address this problem with a *rectangular* belief set. This set permits a separate model choice at every date and history. Each prior is updated separately. The next assumption defines this set.

Assumption 2 (Generation-by-generation ambiguity)**.** *For each state and action, nature's set of one-step-ahead transition laws is the kernel correspondence*

$$\mathcal{P}(x,s,z) = \left\{P_{\rho,z}(\cdot \mid x,s) : \rho \in [\underline{\rho}, \overline{\rho}]\right\}.$$

The set of path priors is the rectangular hull of this correspondence. It contains all path measures generated by measurable selections $\rho_t(h_t) \in [\underline{\rho}, \overline{\rho}]$ from $\mathcal{P}$ at every date and history. These selections can depend on history.

Housing markets, labor markets, and policies can change between generations. The relation between public news and family outcomes can therefore change. Rectangularity permits these changes. It gives dynamic consistency for the maxmin part of the recursion. The Epstein–Schneider theorem does not

cover the added CARA evaluation. I define preferences directly by (6), following recursive robust control (Hansen and Sargent, 2008). Appendix A.7 explains this choice and its entropy representation.

Remark 1 (No learning about informativeness). Under Assumption 2, past signals do not identify the ρ for future signals. Nature can select a new ρ_t at each date and history. This absence of learning is a model assumption. Epstein and Schneider (2007) show that uncertain information quality can prevent belief sets from shrinking after long histories. Learning about signal informativeness remains outside the model. Future work could allow past signals to change beliefs about future informativeness.

The discounting restriction is $\beta\overline{\Phi} < 1$. It replaces the usual restriction $\beta < 1$. Parents with more children assign more total weight to future generations. Their effective discount factor is $\beta\Phi(n)$. A uniform bound below one prevents distant generations from producing unbounded value. It also makes T a contraction. A contraction reduces the distance between candidate value functions.

Proposition 1 (Existence and characterization)**.** *Under Assumption 1, T is a contraction on bounded continuous functions. Its modulus is at most $\beta\overline{\Phi}$. A unique fixed point $V = TV$ therefore exists. Policy functions attain the outer maximum. A worst-case belief $\rho^*(x, s)$ attains the inner minimum.*

Proof. The proof is available in Appendix A.2. □

Neither added evaluation increases distance in the supremum norm. The contraction bound remains $\beta\overline{\Phi}$. Under these conditions, the model has a unique value function. Risk sensitivity and ambiguity can still change optimal choices.

2.4 Equilibrium characterization

The first-order conditions compare current costs with future benefits. They have the usual Barro–Becker structure. The probability weights differ. The parent gives more weight to low-value outcomes when evaluating a plan.

For a given state (x_t, s_t), define the model-ρ certainty equivalent of a plan,

$$K_\rho(n, z; x_t, s_t) \equiv \mathcal{C}_{\gamma,\rho,z}\left[\Phi(n)V(x_{t+1}, s_{t+1}) \mid x_t, s_t\right], \tag{8}$$

and its worst case across models,

$$J(n, z; x_t, s_t) \equiv \min_{\rho\in[\underline{\rho},\overline{\rho}]} K_\rho(n, z; x_t, s_t). \tag{9}$$

Since $u'(c) > 0$, the budget binds at an interior optimum: $c_t^* = m_t - n_t^*(q_t + z_t^*)$. Substitution gives the reduced problem

$$\max_{n,z} \left\{u\big(m_t - n(q_t + z)\big) + \beta J(n, z; x_t, s_t)\right\}. \tag{10}$$

Let

$$\rho_t^* \in \arg\min_{\rho\in[\underline{\rho},\overline{\rho}]} K_\rho(n_t^*, z_t^*; x_t, s_t)$$

be the worst-case informativeness for the chosen plan. The measure Q_t^* starts from $P_{\rho_t^*, z_t^*}$. It assigns new weights proportional to $\exp\{-\gamma\Phi(n_t^*)V(x_{t+1}, s_{t+1})\}$. When $\gamma\Phi(n_t^*) > 0$, low-value states receive more weight. The parameter ρ_t^* selects the worst information model. The CARA evaluation then gives more weight to poor outcomes within that model. These weights describe risk evaluation. They do not change realized outcomes. As $\gamma \to 0$, Q_t^* approaches $P_{\rho_t^*, z_t^*}$. Appendix A.8 gives its Radon–Nikodym derivative and entropy representation.

Assume that the transition has the differentiable shock representation (2). Assume that V is differentiable in the relevant directions. Define the marginal effect of investment on continuation value by

$$\begin{aligned} D_z V_{t+1} \equiv \nabla_x V(x_{t+1}, s_{t+1}) \cdot G_z^x(x_t, s_t, z_t, \varepsilon_{t+1}; \rho_t^*) \\ + V_s(x_{t+1}, s_{t+1})\, G_z^s(x_t, s_t, z_t, \varepsilon_{t+1}; \rho_t^*), \end{aligned} \tag{11}$$

Here G^x and G^s describe the two parts of the next state. $D_z V_{t+1}$ is the extra continuation value from one more unit of investment per child, for each possible future outcome.

Proposition 2 (Interior first-order conditions)**.** *Assume that the transition admits the differentiable shock representation (2). Assume that V is differentiable. Define $D_z V_{t+1}$ by (11). Suppose (c_t^*, n_t^*, z_t^*) is an interior optimum. Suppose the inner minimizer ρ_t^* is unique. Also assume that differentiation under the expectation is valid. Then*

$$\rho_t^* \in \arg\min_{\rho\in[\underline{\rho},\overline{\rho}]} K_\rho(n_t^*, z_t^*; x_t, s_t),$$

and, letting Q_t^* denote the distorted measure associated with (ρ_t^*, n_t^*, z_t^*), the optimal policies satisfy

$$c_t^* = m_t - n_t^*(q_t + z_t^*), \tag{12}$$

$$u'(c_t^*)(q_t + z_t^*) = \beta\Phi'(n_t^*)\mathbb{E}_{Q_t^*}\left[V(x_{t+1}, s_{t+1}) \mid x_t, s_t\right], \tag{13}$$

$$n_t^* u'(c_t^*) = \beta\Phi(n_t^*)\mathbb{E}_{Q_t^*}\left[D_z V_{t+1} \mid x_t, s_t\right]. \tag{14}$$

If $K_\rho(n_t^*, z_t^*; x_t, s_t)$ is differentiable in ρ, the worst-case belief also satisfies the one-dimensional optimality condition

$$\begin{cases} \partial_\rho K_\rho(n_t^*, z_t^*; x_t, s_t)\big|_{\rho=\rho_t^*} = 0, & \rho_t^* \in (\underline{\rho}, \overline{\rho}), \\ \partial_\rho^+ K_\rho(n_t^*, z_t^*; x_t, s_t)\big|_{\rho=\underline{\rho}} \geq 0, & \rho_t^* = \underline{\rho}, \\ \partial_\rho^- K_\rho(n_t^*, z_t^*; x_t, s_t)\big|_{\rho=\overline{\rho}} \leq 0, & \rho_t^* = \overline{\rho}. \end{cases} \tag{15}$$

Proof. The proof is available in Appendix A.3. □

Each condition equates current marginal cost with future marginal benefit. In (13), another child reduces current consumption. The benefit is discounted continuation value with weight $\Phi'(n_t^*)$, evaluated under Q_t^*. In (14), another investment unit per child costs n_t^* units of resources. Each child's value increases by $D_z V_{t+1}$. The total dynastic weight is $\Phi(n_t^*)$.

The worst-case belief depends on the plan. Its derivative does not enter these first-order conditions. With a unique inner minimizer, the envelope theorem permits holding the belief fixed when calculating marginal benefits at the optimum. Ilut and Schneider (2014) use this structure in a business-cycle model.

Divide (13) by (14). Marginal consumption utility cancels. The quantity–quality condition is

$$\frac{q_t + z_t^*}{n_t^*} = \frac{\Phi'(n_t^*)}{\Phi(n_t^*)} \frac{\mathbb{E}_{Q_t^*}[V(x_{t+1}, s_{t+1}) \mid x_t, s_t]}{\mathbb{E}_{Q_t^*}[D_z V_{t+1} \mid x_t, s_t]}. \tag{16}$$

The left side compares two uses of current resources: another child and more investment in each existing child. The right side compares their future benefits, including the different dynastic weights. Both benefits use the same cautious probability measure.

Equation (15) determines worst-case informativeness. At an interior minimum, a small change in ρ has no first-order effect on value. Otherwise, the minimum occurs at a bound. Inertia arises when the interior condition holds over a signal interval.

The probability weights need not cancel from (16). Low continuation value and low investment returns need not occur in the same state. A change in weights can therefore change investment relative to fertility. It can also change fertility itself.

Remark 2 (Special cases). Two limits recover familiar models. As $\gamma \to 0$, the certainty equivalent becomes a conditional expectation (Lemma 1). The result is a risk-neutral robust Barro–Becker problem, with $\Phi(n_t)$ outside the inner minimum. If the ambiguity set instead has only one member, $\underline{\rho} = \overline{\rho} = \rho$, the parent knows the probability law but still uses risk-sensitive preferences. Removing both additions gives the standard Barro–Becker recursion. Appendix A.9 writes out these cases.

3 Signal Informativeness and Fertility Inertia

The recursive model gives optimality conditions. It does not give an explicit fertility rule. I now assume a local normal distribution for continuation value. This assumption shows how worst-case beliefs depend on the plan. Moderately bad news can then lower welfare while fertility, investment, and consumption stay locally unchanged.

3.1 Local normal representation

Around the current state, suppose

$$V(x_{t+1}, s_{t+1}) = \mu(z_t) + \sigma(z_t)\left[\rho s_t + \sqrt{1-\rho^2}\,\varepsilon_{t+1}\right], \qquad \varepsilon_{t+1} \sim N(0,1), \tag{17}$$

with $\sigma(z_t) > 0$.

Equation (17) is a proposed local approximation to continuation value. The functions $\mu(z_t)$ and $\sigma(z_t)$ describe how investment affects its mean and common-risk exposure. I specify these functions directly. I do not derive them from the recursive value function.

Proposition 1 uses bounded value functions. A normal random variable is unbounded. A link between these models requires a different function space with finite CARA values or a valid approximation result. This paper establishes neither link. The local model preserves the economic tradeoff under an additional distributional assumption.

The exercise holds the current budget fixed: $\partial m/\partial s = \partial q/\partial s = 0$. The signal does not enter μ or σ directly. It changes beliefs about future conditions. If news also reduces current income, the budget can change fertility even when the information effect leaves choices unchanged.

The parameter ρ affects the conditional mean and variance: $\mathbb{E}[V(x_{t+1}, s_{t+1}) \mid s_t, \rho] = \mu(z_t) + \sigma(z_t)\rho s_t$ and $\mathrm{Var}(V(x_{t+1}, s_{t+1}) \mid s_t, \rho) = \sigma(z_t)^2(1-\rho^2)$. A larger ρ increases the signal's effect on expected value. It also reduces the remaining variance. Bad news therefore changes the mean and risk in opposing directions.

Now define the dynastic continuation payoff $Y_{t+1} = \Phi(n_t)V(x_{t+1}, s_{t+1})$. Under (17), Y_{t+1} is conditionally normal. The CARA certainty equivalent is therefore

$$K(n_t, z_t; \rho, s_t) = \Phi(n_t)\left[\mu(z_t) + \sigma(z_t)\rho s_t\right] - \frac{\gamma}{2}\Phi(n_t)^2\sigma(z_t)^2(1-\rho^2). \tag{18}$$

The first term is expected dynastic value. The second term is the risk penalty. The factor $\Phi(n_t)^2$ reflects common-risk exposure. More children put more dynastic utility at risk under the same conditions. They do not diversify this common risk.

Using (18), the parent's reduced problem is

$$\max_{n,z}\left\{u\big(m_t - n(q_t + z)\big) + \beta \min_{\rho \in [\underline{\rho}, \overline{\rho}]} K(n, z; \rho, s_t)\right\}. \tag{19}$$

The parent evaluates each plan under every model. She then selects its worst-case value and chooses the best plan. Fixed investment $z = \bar{z}$ gives the two-period example in Section 3.5. Its payoff represents continuation value from the full dynastic model.

3.2 Benchmark with known signal informativeness

First suppose the parent knows ρ. This separates outcome risk from uncertainty about signal informativeness. For a bad signal, a larger ρ lowers expected value and reduces risk. Fertility can rise or fall with ρ. The result depends on which effect is stronger. Proposition 3 states this comparison.

Proposition 3 (Known signal informativeness)**.** *Fix $z_t = z$ and suppose the fertility problem has a unique interior solution $n^*(\rho, s_t)$ satisfying the second-order condition. Assume $\Phi'(n^*) > 0$ and $\sigma(z) > 0$. Under the local normal representation (17),*

$$\operatorname{sgn}\left(\frac{\partial n^*}{\partial \rho}\right) = \operatorname{sgn}\left[s_t + 2\gamma\Phi(n^*)\sigma(z)\rho\right]. \tag{20}$$

Thus higher signal informativeness raises fertility if and only if

$$s_t > -2\gamma\Phi(n^*)\sigma(z)\rho.$$

For endogenous z_t, let H be the objective's Hessian with respect to (n, z) at the optimum. Suppose H is negative definite. Then

$$\begin{pmatrix} \dfrac{\partial n^*}{\partial \rho} \\ \dfrac{\partial z^*}{\partial \rho} \end{pmatrix} = -H^{-1} \begin{pmatrix} \beta\Phi'(n^*)\sigma(z^*)\left[s_t + 2\gamma\Phi(n^*)\sigma(z^*)\rho\right] \\ \beta\Phi(n^*)\sigma_z(z^*)\left[s_t + 2\gamma\Phi(n^*)\sigma(z^*)\rho\right] \end{pmatrix}. \tag{21}$$

Proof. The proof is in Appendix A.4. □

With investment fixed, more informative good news raises expected value and reduces risk. Both effects raise fertility. For bad news, the two effects oppose each other. The risk reduction dominates when $-2\gamma\Phi(n^*)\sigma(z)\rho < s_t < 0$. The expected loss dominates below this interval.

With endogenous investment, both responses in (21) include the same bracketed factor. If this factor is zero, both responses to ρ are locally zero. Next, I allow uncertainty about ρ. The parent then selects the model that balances these effects to give each plan its lowest value.

Figure 1 shows the explicit example derived below. With known ρ, fertility changes smoothly with the signal. Under ambiguity, very bad news uses $\overline{\rho}$. Good news uses $\underline{\rho}$. Fertility stays constant between these cases. The right panel shows the adjustment in the worst-case belief. As the signal rises within the inertia interval, ρ_t^* falls. The worst case assigns less weight to the expected loss and leaves more risk. The family plan stays unchanged until the belief reaches a bound.

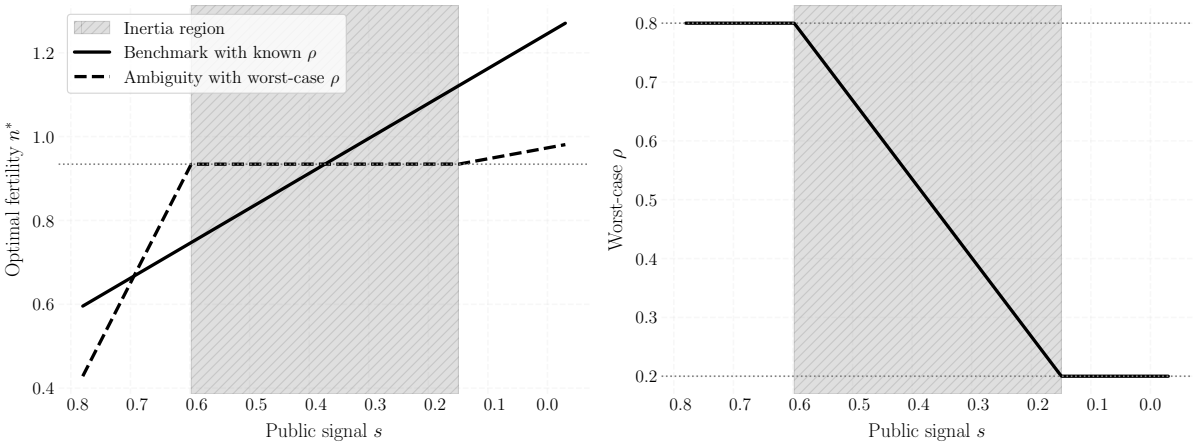


Figure 1: Optimal fertility under known and ambiguous signal informativeness

Notes: The figure compares fertility under known informativeness and ambiguity. Under ambiguity, the parent uses the worst admissible model. Parameters are $u(c) = c$, $\Phi(n) = n$, $y = 10$, $q = 1$, $\beta = 0.95$, $\mu = 1.8$, $\sigma = 1$, $\gamma = 0.8$, $\underline{\rho} = 0.2$, and $\overline{\rho} = 0.8$. The known-ρ case uses $\rho = 0.5$. Fertility within the inertia interval is $n^0 = 0.934$. The interval is approximately $s \in [-0.598, -0.149]$.

3.3 Ambiguous signal informativeness

Now suppose signal informativeness is uncertain. Fix a plan (n, z) with $\Phi(n) > 0$. At $n = 0$, continuation value is zero and does not depend on ρ. No model selection is necessary for that plan. For a positive family size, the parent evaluates $\min_{\rho \in [\underline{\rho}, \overline{\rho}]} K(n, z; \rho, s_t)$. Using (18), the derivative with respect to ρ is

$$K_\rho(n, z; \rho, s_t) = \Phi(n)\sigma(z)s_t + \gamma\Phi(n)^2\sigma(z)^2\rho,$$

and

$$K_{\rho\rho}(n, z; \rho, s_t) = \gamma\Phi(n)^2\sigma(z)^2 > 0.$$

Because K is strictly convex in ρ, the worst case is unique. First find the minimum without the bounds. If it falls outside the allowed range, use the nearest endpoint:

$$\rho^w(n, z, s_t) = \text{Proj}_{[\underline{\rho}, \overline{\rho}]} \left(-\frac{s_t}{\gamma\Phi(n)\sigma(z)} \right). \tag{22}$$

Suppose $0 \leq \underline{\rho} < \overline{\rho}$. For good news ($s_t \geq 0$), the unconstrained minimizer is nonpositive. The constrained minimum therefore uses $\rho^w = \underline{\rho}$. The parent gives good news the lowest admissible informativeness. For $s_t \leq -\gamma\Phi(n)\sigma(z)\overline{\rho}$, the minimum uses $\rho^w = \overline{\rho}$. The parent evaluates sufficiently bad news at the highest informativeness.

Between these two cases, for

$$-\gamma\Phi(n)\sigma(z)\overline{\rho} < s_t < -\gamma\Phi(n)\sigma(z)\underline{\rho},$$

the worst case is interior, so

$$\rho^w(n, z, s_t) = -\frac{s_t}{\gamma\Phi(n)\sigma(z)}.$$

Within this interval, the worst-case belief can change without reaching a bound. It depends on the signal and the plan. A larger family or an investment that raises $\sigma(z)$ increases common-risk exposure. This changes the model that is worst for the plan. The next proposition shows when this adjustment leaves choices unchanged.

Proposition 4 (Fertility inertia)**.** *Let (n^0, z^0) be the unique interior maximizer of the signal-free problem*

$$\max_{n,z} \left\{ u\big(m_t - n(q_t + z)\big) + \beta \left[\Phi(n)\mu(z) - \frac{\gamma}{2}\Phi(n)^2\sigma(z)^2\right] \right\}. \tag{23}$$

Suppose the objective in (23) is strictly concave in a neighborhood of (n^0, z^0), with $\Phi(n^0) > 0$ and $\sigma(z^0) > 0$. If the signal satisfies

$$-\gamma\Phi(n^0)\sigma(z^0)\overline{\rho} < s_t < -\gamma\Phi(n^0)\sigma(z^0)\underline{\rho}, \tag{24}$$

then the worst-case signal informativeness is interior at (n^0, z^0), and the optimal policies in (19) satisfy, locally,

$$n^*(s_t) = n^0, \qquad z^*(s_t) = z^0, \qquad c^*(s_t) = m_t - n^0(q_t + z^0). \tag{25}$$

Thus, throughout the inertia interval, optimal fertility, child investment, and consumption are locally independent of the public signal.

Proof. The proof is available in Appendix A.5. □

This result is local. It excludes a better nearby plan with an interior worst-case belief. A global result requires comparisons with more distant plans. These include childless plans and plans with boundary worst-case beliefs. Such plans need not have the same signal loss. The appendix states additional conditions for a global optimum.

The endpoints of (24) require a separate check. At an endpoint, the belief reaches a bound. A nearby plan can then receive an additional convex term in value. The appendix gives a stronger second-order condition that preserves local optimality. The example in Section 3.5 satisfies this condition automatically.

3.4 The welfare cost of ambiguous news

Substituting the interior worst-case belief from (22) into (18) shows how the effect of news enters value:

$$\min_{\rho\in[\underline{\rho},\overline{\rho}]} K(n, z; \rho, s_t) = \Phi(n)\mu(z) - \frac{\gamma}{2}\Phi(n)^2\sigma(z)^2 - \frac{s_t^2}{2\gamma}. \tag{26}$$

The first two terms give expected dynastic value less the full common-risk penalty. The last term is the signal loss. For plans with interior worst-case beliefs, this loss does not depend on fertility or investment. Consider a parent comparing nearby plans with different fertility and investment. Worse news subtracts the same value from each plan. It does not change their relative value. The marginal conditions therefore match the signal-free problem, although welfare falls.

Corollary 1 (Width of the inertia interval)**.** *For a fixed signal-free optimum (n^0, z^0), the length of the fertility-inertia interval is*

$$\gamma\, \Phi(n^0)\, \sigma(z^0)\, (\overline{\rho} - \underline{\rho}). \tag{27}$$

For a fixed plan, width increases with risk sensitivity γ, fertility exposure $\Phi(n^0)$, continuation-value exposure $\sigma(z^0)$, and the informativeness range $\overline{\rho} - \underline{\rho}$.

Equation (27) gives the role of each factor. A larger γ makes remaining risk more costly. Higher $\Phi(n^0)$ or $\sigma(z^0)$ increases exposure. A wider informativeness interval allows a larger change in the worst-case belief before it reaches a bound. Each change widens the inertia interval for a fixed signal-free plan. If a parameter also changes n^0 or z^0, the total effect must include those changes. The inertia interval lies between the lower-informativeness evaluation of good news and the higher-informativeness evaluation of very bad news.

3.5 Closed-form special case

The parent can so far change fertility and investment. For an explicit fertility rule, I now fix investment in a two-period model. Set $u(c) = c$, $\Phi(n) = n$, and $z = 0$. Then

$$\mu \equiv \mu(0), \qquad \sigma \equiv \sigma(0), \qquad q_t = q, \qquad y \equiv m_t.$$

Here q denotes the resource cost of a child and y the parent's current resources.

In the absence of a signal, fertility solves

$$\max_{0 \le n \le y/q} \left\{ y - qn + \beta \left[n\mu - \frac{\gamma}{2} n^2 \sigma^2 \right] \right\}.$$

The objective is strictly concave because its second derivative with respect to n is negative:

$$\frac{\partial^2}{\partial n^2} \left\{ y - qn + \beta \left[n\mu - \frac{\gamma}{2} n^2 \sigma^2 \right] \right\} = -\beta\gamma\sigma^2 < 0.$$

The interior first-order condition therefore gives

$$n^0 = \frac{\mu - q/\beta}{\gamma\sigma^2}. \tag{28}$$

This solution is interior when

$$\mu > q/\beta \qquad \text{and} \qquad qn^0 < y.$$

The first inequality says that a child's expected dynastic payoff must exceed its cost after discounting. The second says that the family can afford the chosen number of children and still consume something.

Define

$$\lambda \equiv \frac{\mu - q/\beta}{\sigma}.$$

The numerator of λ is the mean payoff less the discount-adjusted cost. Division by σ gives the gain per unit of common risk. Thus λ measures the appeal of parenthood relative to risk. Equation (28) gives

$$\gamma\sigma n^0 = \lambda.$$

Therefore the inertia interval becomes

$$s_t \in [-\lambda\overline{\rho},\, -\lambda\underline{\rho}], \qquad \text{with width} \qquad \lambda(\overline{\rho} - \underline{\rho}). \tag{29}$$

The width in (29) has two factors. The factor λ determines optimal common-risk exposure. The factor $\overline{\rho} - \underline{\rho}$ measures uncertainty about signal informativeness. When ρ is known, the worst-case belief cannot adjust. The inertia interval then has zero width. An expansion around a fixed midpoint increases width in direct proportion. Low informativeness alone does not produce this interval. Informativeness must be uncertain.

The model also gives an explicit solution outside the inertia interval when $0 < n^*(s_t) < y/q$. Sufficiently bad news leads to zero fertility. If $\underline{\rho} > 0$, sufficiently good news makes the budget bind. At these corners, constraints keep fertility unchanged. For an interior fertility choice and boundary worst-case belief $\hat{\rho} \in \{\underline{\rho}, \overline{\rho}\}$, the solution is

$$n^*(s_t; \hat{\rho}) = \frac{\mu + \sigma\hat{\rho}s_t - q/\beta}{\gamma\sigma^2(1 - \hat{\rho}^2)}.$$

Corollary 2 (Closed-form policy function)**.** *Assume the specification of this subsection with* $0 \le \underline{\rho} < \overline{\rho} < 1$*, and suppose the solution remains interior,* $0 < n^*(s_t) < y/q$*. Then the ambiguous-signal policy is*

$$n^*(s_t) = \begin{cases} \dfrac{\mu + \sigma\overline{\rho}s_t - q/\beta}{\gamma\sigma^2(1 - \overline{\rho}^2)}, & s_t \le -\lambda\overline{\rho}, \\[2ex] n^0, & -\lambda\overline{\rho} \le s_t \le -\lambda\underline{\rho}, \\[2ex] \dfrac{\mu + \sigma\underline{\rho}s_t - q/\beta}{\gamma\sigma^2(1 - \underline{\rho}^2)}, & s_t \ge -\lambda\underline{\rho}. \end{cases} \tag{30}$$

The three pieces join continuously. The policy is flat within the inertia interval, and outside the region its slope is

$$\frac{\partial n^*(s_t;\hat{\rho})}{\partial s_t} = \frac{\hat{\rho}}{\gamma\sigma(1-\hat{\rho}^2)},$$

with $\hat{\rho} = \overline{\rho}$ on the bad-news side and $\hat{\rho} = \underline{\rho}$ on the good-news side.

The three parts are continuous, but their slopes differ. The bad-news part reaches n^0 at $s_t = -\lambda\overline{\rho}$. The good-news part reaches n^0 at $s_t = -\lambda\underline{\rho}$. The ratio $\rho/(1-\rho^2)$ increases on $[0,1)$. The slope is therefore greater on the $\overline{\rho}$ side. Parents reduce fertility more strongly for very bad news than they increase it for good news.

3.6 Generosity and legibility of family policy

The previous results vary news and hold the current budget fixed. I now study a benefit that reduces child costs. A household can respond to reliable support while keeping the same plan after a change in news.

Let $\tau \geq 0$ be the promised benefit per child. The household may receive less. It can fail an income test or find no available childcare place. Application costs or a rule change can also reduce the amount received. Let α be the *delivery rate*: the share of the promised benefit that reaches the household. The parent knows an interval for this rate. She assigns no probabilities within the interval:

$$\alpha \in [\underline{\alpha}, \overline{\alpha}] \subseteq [0,1], \tag{31}$$

The delivery rate is now ambiguous. A more legible program has a higher lowest delivery rate $\underline{\alpha}$. Reducing only $\overline{\alpha}$ does not change choices in this model. The interval $[\underline{\rho}, \overline{\rho}]$ separately measures uncertainty about signal informativeness. The net cost per child is $q - \alpha\tau$. The parent from Section 3.5 solves

$$\max_{0\leq n\leq y/(q-\underline{\alpha}\tau)} \min_{\substack{\rho\in[\underline{\rho},\overline{\rho}]\\ \alpha\in[\underline{\alpha},\overline{\alpha}]}} \left\{ y - (q-\alpha\tau)n + \beta\left[n(\mu + \sigma\rho s_t) - \frac{\gamma}{2}n^2\sigma^2(1-\rho^2)\right]\right\}. \tag{32}$$

For $n > 0$ and $\tau > 0$, a higher delivery rate increases value. The worst case therefore uses $\alpha = \underline{\alpha}$ for every signal and plan. The minimizations over delivery and informativeness separate. The previous results apply after replacing q with

$$q_e(\tau) \equiv q - \underline{\alpha}\tau.$$

Proposition 5 (Generosity and legibility)**.** *Consider problem (32) with $0 \leq \underline{\alpha} \leq \overline{\alpha} \leq 1$ and $q - \underline{\alpha}\tau > 0$, and suppose the signal-free solution is interior. Then the following hold.*

(i) The lowest delivery rate is $\alpha^* = \underline{\alpha}$ for every plan with $n > 0$, and all results of Sections 3.1–3.5

hold with q replaced by $q_e(\tau)$. In particular, inside the inertia region fertility equals

$$n^0(\tau) = \frac{\mu - \big(q - \underline{\alpha}\tau\big)/\beta}{\gamma\sigma^2}. \tag{33}$$

(ii) The fertility response to the subsidy is

$$\frac{\partial n^0}{\partial \tau} = \frac{\underline{\alpha}}{\beta\gamma\sigma^2}. \tag{34}$$

The lowest delivery rate $\underline{\alpha}$ scales the marginal response.

(iii) (Legibility equivalence.) The uncertain program is $(\tau, [\underline{\alpha}, \overline{\alpha}])$. It has the same effect on every household choice as a certain benefit of $\underline{\alpha}\tau$ with known $\alpha \equiv 1$. For $\underline{\alpha} > 0$, matching a certain benefit τ^* requires a promise larger by the factor $1/\underline{\alpha}$. At $\underline{\alpha} = 0$, no finite promise can match a positive certain benefit.

(iv) The width of the inertia interval, $\lambda(\tau)(\overline{\rho} - \underline{\rho})$ with $\lambda(\tau) = \big[\mu - (q - \underline{\alpha}\tau)/\beta\big]/\sigma$, is nondecreasing in τ,

$$\frac{\partial}{\partial \tau}\Big[\lambda(\tau)(\overline{\rho} - \underline{\rho})\Big] = \frac{\underline{\alpha}}{\beta\sigma}(\overline{\rho} - \underline{\rho}) \geq 0,$$

with strict inequality if and only if $\underline{\alpha} > 0$ and $\overline{\rho} > \underline{\rho}$.

Proof. The proof is available in Appendix A.6. □

Part (ii) gives the limit as $\underline{\alpha} \to 0$. The fertility effect of an additional promised unit approaches zero. Government spending can still be positive. Let the realized delivery rate be $\alpha_0 \in [\underline{\alpha}, \overline{\alpha}]$ with $\alpha_0 > 0$. The payment per child is $g = \alpha_0\tau$. The fertility response per payment unit is

$$\frac{\partial n^0}{\partial g} = \frac{\underline{\alpha}}{\alpha_0} \cdot \frac{1}{\beta\gamma\sigma^2} \;\leq\; \frac{1}{\beta\gamma\sigma^2},$$

This equals the response to a certain transfer multiplied by $\underline{\alpha}/\alpha_0 \leq 1$. Parents use the lowest delivery rate. The government pays according to the realized delivery rate.

For example, suppose the benefit is 100 units per child. Parents consider delivery of at least one half certain. The rest is uncertain. They make the same choice as under a certain payment of 50 units. If the government later pays 100, the extra payment was not reliable support at the time of choice. The ratio of $\underline{\alpha}$ to α_0 measures this difference. The studies in the introduction do not estimate these bounds.

Part (iv) concerns the response to later news. More effective support raises fertility and common-risk exposure. It shifts and widens the inertia interval. Because the interval also shifts, a greater width does not imply a smaller response at every signal.

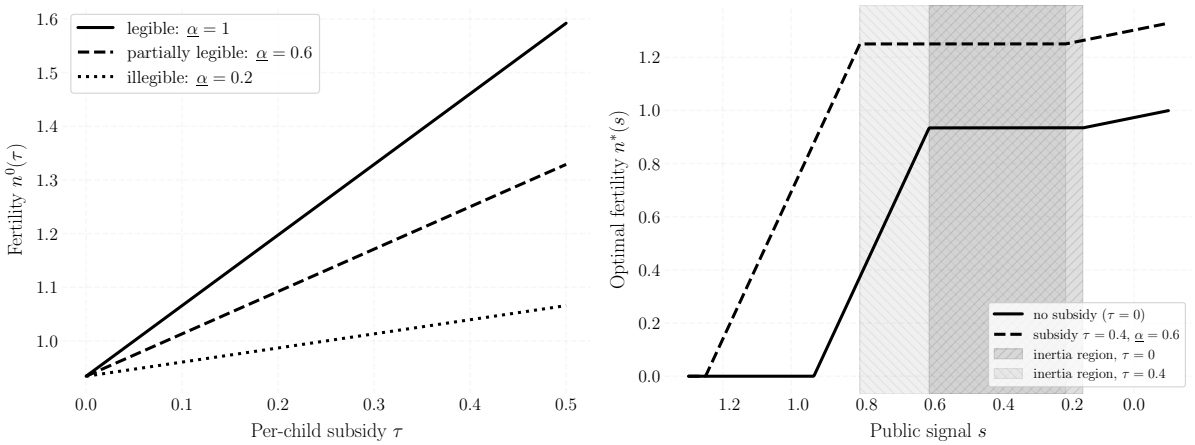


Figure 2: Generosity versus legibility of family policy

Notes: The left panel shows $n^0(\tau)$ from (33) against the promised benefit τ. It uses three lowest delivery rates $\underline{\alpha}$. The slope is $\underline{\alpha}/(\beta\gamma\sigma^2)$. A higher lowest delivery rate therefore gives a larger fertility response. The right panel compares $n^*(s)$ without a subsidy and with $\tau = 0.4$. The subsidy has $\underline{\alpha} = 0.6$. It raises fertility and shifts and widens the inertia interval (Proposition 5(iv)). Parameters are $u(c) = c$, $\Phi(n) = n$, $y = 10$, $q = 1$, $\beta = 0.95$, $\mu = 1.8$, $\sigma = 1$, $\gamma = 0.8$, $\underline{\rho} = 0.2$, and $\overline{\rho} = 0.8$.

An announcement can change reliable support and beliefs about future conditions. To isolate its information effect, hold effective support fixed. Corollary 2 evaluates good news at $\underline{\rho}$. The fertility slope is $\underline{\rho}/[\gamma\sigma(1-\underline{\rho}^2)]$. Within $[-\lambda\overline{\rho}, -\lambda\underline{\rho}]$, the information effect leaves fertility unchanged. If reliable support also increases, it has a separate incentive effect. An observed response to an announcement can include both effects.

A reform can improve delivery, information, or both. More reliable delivery raises $\underline{\alpha}$ toward $\overline{\alpha}$. Less uncertainty about informativeness narrows $[\underline{\rho}, \overline{\rho}]$. These changes affect width $W = \lambda(\tau, \underline{\alpha})(\overline{\rho} - \underline{\rho})$ in opposite directions. A higher lowest delivery rate raises effective support. For $\tau > 0$, it also increases fertility and common-risk exposure. Proposition 5(iv) gives $\partial W/\partial\underline{\alpha} = \tau(\overline{\rho} - \underline{\rho})/(\beta\sigma) \geq 0$. A narrower informativeness interval reduces width: $\partial W/\partial(\overline{\rho} - \underline{\rho}) = \lambda > 0$. The total change is

$$dW = \frac{\tau(\overline{\rho} - \underline{\rho})}{\beta\sigma}\, d\underline{\alpha} + \lambda\, d(\overline{\rho} - \underline{\rho}).$$

Width falls only if the reduction in $\overline{\rho} - \underline{\rho}$ exceeds the exposure effect from better delivery. Reliable payments encourage births. Clearer information restricts changes in worst-case beliefs. A reform can affect both. Figure 2 shows the delivery effect.

Guaranteed childcare places and clear eligibility rules can change parents' delivery bounds. Enforceable return-to-work rights and credible benefit schedules can also change those bounds. The promised amount τ can change separately. Under the model's assumptions, a smaller certain benefit can have the same fertility effect as a larger uncertain benefit. Their relative government cost depends on realized delivery.

3.7 Scope and limitations

The exact inertia result depends on specific assumptions. Parents evaluate each plan by its worst-case value. Fertility is a continuous choice. The analysis also uses a local normal representation of continuation value.

Smooth-ambiguity models instead assign weight to several possible models (Klibanoff et al., 2005, 2009). This paper does not solve the household problem under those preferences. It therefore does not establish exact inertia or a near-zero response to news in that case.

Actual family size is a whole number. This paper does not solve the model with integer fertility. Discrete choices can remain unchanged over a range of signals even without ambiguity. Such inaction alone would not identify this paper's mechanism. A comparison must account for the childless option and switches between family sizes.

The recursive model does not establish the local normal representation. I therefore treat exact inertia as a result for this benchmark, not as a general prediction. Future work must test whether the mechanism persists with other preferences, integer fertility, and other continuation-value distributions.

An empirical test could examine parents who report a worse outlook and keep the same family plan. Such a test needs a link between survey responses and model welfare. Fertility intentions alone do not measure welfare.

4 Conclusion

Parents can process bad news and keep the same family plan. In the local normal model, uncertain informativeness allows the worst-case belief to adjust. For moderately bad signals, nearby plans with interior worst-case beliefs have the same signal loss. Welfare falls, but fertility, consumption, and investment stay locally unchanged. The explicit model places this inertia interval between stronger responses to very bad news and weaker responses to good news.

Reliable transfers can still change fertility. Parents use the lowest delivery rate they consider possible. The government can later deliver more. Higher effective support raises fertility. It can also widen the inertia interval through higher common-risk exposure. Less uncertainty about informativeness restricts the adjustment of worst-case beliefs. The model therefore separates three policy features: the promised benefit, reliable delivery, and information about future conditions.

The exact inertia result has the limits stated in Section 3.7. The paper does not establish the result under smooth ambiguity or integer fertility. The evidence in the introduction motivates the model. It does not measure the mechanism. Future research could connect the local normal assumption to the recursive value function. It could also allow learning about informativeness. Empirical tests must separate news from changes in current resources. They must measure delivery uncertainty separately from the

promised benefit.

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Data availability This theoretical article uses no proprietary data. The model generates all figures. The author can provide replication code on request.

Materials availability Not applicable.

Code availability The author can provide the code for figures and replication on request.

Author contribution HongSeok Kim is the sole author. He completed all aspects of this work.

Appendices

A Proofs

A.1 Derivation of CARA

Let X_{t+1} denote the future net payoff associated with a child. Its conditional mean and variance are

$$\mathbb{E}[X_{t+1}|s,\rho] = \mu + \sigma\rho s$$

and

$$\mathrm{Var}(X_{t+1} \mid s,\rho) = \sigma^2(1-\rho^2).$$

For a parent with n children, the total dynastic payoff is

$$Z_{t+1} = \Phi(n)X_{t+1},$$

where $\Phi(n)$ is the dynastic weight. Conditional on the signal s and informativeness parameter ρ, this payoff is normally distributed:

$$Z_{t+1}|s,\rho \sim N(m,v),$$

where

$$m = \Phi(n)[\mu + \sigma\rho s]$$

and

$$v = \Phi(n)^2\sigma^2(1-\rho^2).$$

To evaluate this risky payoff, write Z for Z_{t+1} and use the CARA utility function

$$U(Z) = -\exp(-\gamma Z),$$

where $\gamma > 0$ is the coefficient of absolute risk aversion. The certainty equivalent CE is defined by

$$-\exp(-\gamma CE) = \mathbb{E}(-\exp(-\gamma Z)).$$

Since $Z \sim N(m,v)$, the normal moment-generating function gives

$$\mathbb{E}\left[\exp(-\gamma Z)\right] = \exp\left(-\gamma m + \frac{1}{2}\gamma^2 v\right).$$

Substituting this expression into the definition of the certainty equivalent yields

$$\exp(-\gamma CE) = \exp\left(-\gamma m + \frac{1}{2}\gamma^2 v\right).$$

Taking logarithms and rearranging gives

$$CE = m - \frac{\gamma}{2}v = \mathbb{E}[Z] - \frac{\gamma}{2}\operatorname{Var}(Z).$$

Finally, substituting the conditional mean and variance of Z gives

$$CE(n; s, \rho) = \Phi(n)[\mu + \sigma\rho s] - \frac{\gamma}{2}\Phi(n)^2\sigma^2(1-\rho^2).$$

A.2 Proof of Proposition 1

Proof. Let $C_b(\mathcal{X}\times\mathcal{S})$ be the space of bounded continuous functions on the state space. Use the supremum norm $\|\cdot\|_\infty$. This is a complete metric space. Write a feasible action as $a = (c, n, z) \in \Gamma(x, s)$. Let $V' = V(x', s')$. First, I show that T maps C_b into itself. I then establish contraction with modulus $\beta\overline{\Phi}$ and use measurable selections to characterize the fixed point.

Two properties of $\mathcal{C}_{\gamma,\rho,z}$ give the required bound. For $\gamma > 0$, both follow from $\mathcal{C}_{\gamma,\rho,z}[Y] = -\gamma^{-1}\log\mathbb{E}_{\rho,z}[e^{-\gamma Y}]$. At $\gamma = 0$, they follow from conditional expectation. Monotonicity preserves payoff order: $Y_1 \le Y_2 \Rightarrow \mathcal{C}_{\gamma,\rho,z}[Y_1] \le \mathcal{C}_{\gamma,\rho,z}[Y_2]$. Translation invariance preserves certain additions: $\mathcal{C}_{\gamma,\rho,z}[Y+k] = \mathcal{C}_{\gamma,\rho,z}[Y] + k$ for constant k. These properties bound the change in the certainty equivalent. If $\|Y_1 - Y_2\|_\infty \le \eta$, then

$$\left|\mathcal{C}_{\gamma,\rho,z}[Y_1] - \mathcal{C}_{\gamma,\rho,z}[Y_2]\right| \le \eta. \tag{$*$}$$

T maps C_b into C_b. Fix $V \in C_b$. Assumption 1 makes Γ nonempty, compact-valued, and continuous. The interval $[\underline{\rho}, \overline{\rho}]$ is compact, u is continuous and bounded on the graph of Γ ($|u| \le \overline{u}$), and $0 \le \Phi(n) \le \overline{\Phi}$. Thus $|\Phi(n)V'| \le \overline{\Phi}\|V\|_\infty$. Applying $(*)$ to the constant bounds gives $|\mathcal{C}_{\gamma,\rho,z}[\Phi(n)V']| \le \overline{\Phi}\|V\|_\infty$ for every ρ. It follows that $|(TV)(x,s)| \le \overline{u} + \beta\overline{\Phi}\|V\|_\infty$, so TV is bounded.

Next, define $H_V(x, s, a, \rho) = u(x, s, a) + \beta\mathcal{C}_{\gamma,\rho,z}[\Phi(n)V']$. Assumption 1 makes the transition law continuous. The integrand $e^{-\gamma\Phi(n)V'}$ is bounded and continuous. The integral is therefore continuous in (x, s, a, ρ). Thus H_V is continuous. The interval for ρ is compact and independent of (x, s, a). Berge's theorem makes $G_V(x, s, a) := \min_\rho H_V(x, s, a, \rho)$ continuous. Its argmin correspondence is nonempty and compact-valued. Apply the same theorem to Γ. Then $(TV)(x, s) = \max_{a\in\Gamma(x,s)} G_V(x, s, a)$ is also continuous.

T is a contraction. Let $V_1, V_2 \in C_b$ and $\Delta := \|V_1 - V_2\|_\infty$. For fixed feasible (x, s, a, ρ), $|V_1' - V_2'| \le \Delta$

and $\Phi(n) \geq 0$ give $\|\Phi(n)V_1' - \Phi(n)V_2'\|_\infty \leq \overline{\Phi}\Delta$, so by $(*)$,

$$\left|\mathcal{C}_{\gamma,\rho,z}[\Phi(n)V_1'] - \mathcal{C}_{\gamma,\rho,z}[\Phi(n)V_2']\right| \leq \overline{\Phi}\Delta$$

uniformly in ρ. Neither minimization nor maximization increases this bound. Suppose $\sup_\rho |f_\rho - g_\rho| \leq \eta$. Then $f \leq g + \eta$. Take minima on both sides and repeat with f and g interchanged. This gives $|\min_\rho f_\rho - \min_\rho g_\rho| \leq \eta$. The same argument applies to maxima. Current utility $u(x, s, a)$ cancels from the Bellman value difference. Apply the bound to the inner minimum. Multiply it by β, then apply it to the outer maximum. For every (x, s),

$$|(TV_1)(x, s) - (TV_2)(x, s)| \leq \beta\overline{\Phi}\Delta,$$

hence $\|TV_1 - TV_2\|_\infty \leq \beta\overline{\Phi}\|V_1 - V_2\|_\infty$. Since $\beta\overline{\Phi} < 1$ by Assumption 1, T is a contraction on C_b, and Banach's fixed-point theorem gives a unique $V \in C_b$ with $V = TV$.

Characterization. At this fixed point, define

$$\begin{aligned}\mathcal{R}_V(x, s, a) &:= \arg\min_{\rho\in[\underline{\rho},\overline{\rho}]} \mathcal{C}_{\gamma,\rho,z}\left[\Phi(n)V'\right],\\ \mathcal{P}_V(x, s) &:= \arg\max_{a\in\Gamma(x,s)} \left\{u(x, s, a) + \beta \min_\rho \mathcal{C}_{\gamma,\rho,z}\left[\Phi(n)V'\right]\right\}.\end{aligned}$$

Berge's theorem makes both correspondences nonempty and compact-valued. Each admits a measurable selection. Select $a^*(x, s) = (c^*, n^*, z^*)(x, s) \in \mathcal{P}_V(x, s)$ and $\rho^*(x, s) \in \mathcal{R}_V(x, s, a^*(x, s))$. Then

$$V(x, s) = u\big(x, s, a^*(x, s)\big) + \beta\, \mathcal{C}_{\gamma,\rho^*(x,s),z^*(x,s)}\left[\Phi(n^*(x, s))V'\right] \qquad \text{for every } (x, s).$$

The worst-case belief $\rho^*(x, s)$ is the inner minimizer evaluated at the chosen policy $a^*(x, s)$. This dependence of the worst-case belief on the chosen plan is central to the inertia analysis. □

A.3 Proof of Proposition 2

Proof. Fix the current state (x_t, s_t) and suppress it where no confusion can arise. Let

$$J(n, z; x_t, s_t) := \min_{\rho\in[\underline{\rho},\overline{\rho}]} K_\rho(n, z; x_t, s_t).$$

The parent's problem can be written as

$$\max_{c,n,z} \{u(c) + \beta J(n, z; x_t, s_t)\}$$

subject to

$$c + n(q_t + z) \leq m_t.$$

Since $u'(c) > 0$, the budget constraint binds at an optimum. Hence

$$c_t^* = m_t - n_t^*(q_t + z_t^*),$$

which gives (12).

Let λ_t denote the multiplier on the budget constraint. For an interior solution, the first-order conditions are

$$u'(c_t^*) = \lambda_t,$$

$$\beta J_n(n_t^*, z_t^*; x_t, s_t) = \lambda_t(q_t + z_t^*),$$

and

$$\beta J_z(n_t^*, z_t^*; x_t, s_t) = \lambda_t n_t^*.$$

Substituting $\lambda_t = u'(c_t^*)$ gives

$$u'(c_t^*)(q_t + z_t^*) = \beta J_n(n_t^*, z_t^*; x_t, s_t)$$

and

$$n_t^* u'(c_t^*) = \beta J_z(n_t^*, z_t^*; x_t, s_t).$$

It remains to compute the derivatives of J. For fixed ρ, use the differentiable shock representation

$$x_{t+1} = G^x(x_t, s_t, z_t, \varepsilon_{t+1}; \rho), \qquad s_{t+1} = G^s(x_t, s_t, z_t, \varepsilon_{t+1}; \rho),$$

The conditional law of ε_{t+1} can depend on ρ. It does not depend on n or z (Assumption 1(v)). Investment changes the next state for a given shock. It does not change that shock's probability. The derivative therefore has no likelihood-score term. If the shock density were $p_{\rho,z}$, the investment derivative would also include $-\gamma^{-1}\mathbb{E}_Q[\partial_z \log p_{\rho,z}]$. Define

$$Y_{t+1}(n, z, \rho) := \Phi(n)V(G^x(x_t, s_t, z, \varepsilon_{t+1}; \rho), G^s(x_t, s_t, z, \varepsilon_{t+1}; \rho)).$$

For $\gamma > 0$,

$$K_\rho(n, z; x_t, s_t) = -\frac{1}{\gamma} \log \mathbb{E}_\rho \left[\exp\{-\gamma Y_{t+1}(n, z, \rho)\} \mid x_t, s_t\right],$$

where the expectation is over the shock ε_{t+1}.

The differentiability and domination assumptions permit differentiation under the expectation. For a

control $a \in \{n, z\}$,

$$\frac{\partial K_\rho}{\partial a} = -\frac{1}{\gamma} \frac{\mathbb{E}_\rho \left[-\gamma \exp\{-\gamma Y_{t+1}\} \frac{\partial Y_{t+1}}{\partial a} \mid x_t, s_t \right]}{\mathbb{E}_\rho \left[\exp\{-\gamma Y_{t+1}\} \mid x_t, s_t \right]} = \frac{\mathbb{E}_\rho \left[\exp\{-\gamma Y_{t+1}\} \frac{\partial Y_{t+1}}{\partial a} \mid x_t, s_t \right]}{\mathbb{E}_\rho \left[\exp\{-\gamma Y_{t+1}\} \mid x_t, s_t \right]}.$$

Define the distorted measure $Q_{\rho,n,z}$ by

$$\frac{\mathrm{d}Q_{\rho,n,z}}{\mathrm{d}P_{\rho,z}} = \frac{\exp\{-\gamma Y_{t+1}(n, z, \rho)\}}{\mathbb{E}_{\rho,z} \left[\exp\{-\gamma Y_{t+1}(n, z, \rho)\} \mid x_t, s_t \right]}.$$

In the shock representation, the same formula gives more weight to shocks that lead to lower payoffs. This exponential reweighting gives

$$\frac{\partial K_\rho}{\partial a} = \mathbb{E}_{Q_{\rho,n,z}} \left[\frac{\partial Y_{t+1}}{\partial a} \mid x_t, s_t \right].$$

Since the inner minimizer ρ_t^* is unique, Danskin's envelope theorem (Danskin, 1966; Milgrom and Segal, 2002) implies

$$J_a(n_t^*, z_t^*; x_t, s_t) = \left. \frac{\partial K_\rho(n_t^*, z_t^*; x_t, s_t)}{\partial a} \right|_{\rho=\rho_t^*}, \qquad a \in \{n, z\}.$$

Thus there is no additional term involving the derivative of ρ_t^* with respect to n or z.

Now compute the two derivatives of Y_{t+1}. Since the transition does not depend directly on n,

$$\frac{\partial Y_{t+1}}{\partial n} = \Phi'(n_t^*) V(x_{t+1}, s_{t+1}).$$

Therefore

$$J_n(n_t^*, z_t^*; x_t, s_t) = \Phi'(n_t^*) \mathbb{E}_{Q_t^*} \left[V(x_{t+1}, s_{t+1}) \mid x_t, s_t \right],$$

where $Q_t^* = Q_{\rho_t^*, n_t^*, z_t^*}$. Substituting into the first-order condition for n gives

$$u'(c_t^*)(q_t + z_t^*) = \beta \Phi'(n_t^*) \mathbb{E}_{Q_t^*} \left[V(x_{t+1}, s_{t+1}) \mid x_t, s_t \right],$$

which is (13).

For z, the chain rule gives

$$\begin{aligned} \frac{\partial Y_{t+1}}{\partial z} &= \Phi(n_t^*) \left[\nabla_x V(x_{t+1}, s_{t+1}) \cdot G_z^x(x_t, s_t, z_t^*, \varepsilon_{t+1}; \rho_t^*) \right. \\ &\qquad \left. + V_s(x_{t+1}, s_{t+1}) G_z^s(x_t, s_t, z_t^*, \varepsilon_{t+1}; \rho_t^*) \right] \\ &= \Phi(n_t^*) D_z V_{t+1}. \end{aligned}$$

Hence

$$J_z(n_t^*, z_t^*; x_t, s_t) = \Phi(n_t^*)\mathbb{E}_{Q_t^*}\left[D_z V_{t+1} \mid x_t, s_t\right].$$

Substituting into the first-order condition for z gives

$$n_t^* u'(c_t^*) = \beta\Phi(n_t^*)\mathbb{E}_{Q_t^*}\left[D_z V_{t+1} \mid x_t, s_t\right],$$

which is (14).

Finally, by definition of the inner problem,

$$\rho_t^* \in \arg\min_{\rho\in[\underline{\rho},\overline{\rho}]} K_\rho(n_t^*, z_t^*; x_t, s_t).$$

If K_ρ is differentiable and the minimum is interior, the first-order condition is $\partial_\rho K_\rho|_{\rho=\rho_t^*} = 0$. At $\rho_t^* = \underline{\rho}$, the right derivative satisfies $\partial_\rho^+ K_\rho|_{\rho=\underline{\rho}} \geq 0$. At $\rho_t^* = \overline{\rho}$, the left derivative satisfies $\partial_\rho^- K_\rho|_{\rho=\overline{\rho}} \leq 0$. These conditions give (15) and complete the proof. □

A.4 Proof of Proposition 3

Proof. Fix $z_t = z$ and write

$$c(n) = m_t - n(q_t + z).$$

The fertility first-order condition is

$$G(n, \rho; s_t) \equiv -(q_t + z)u'(c(n)) + \beta\Phi'(n)\left[\mu(z) + \sigma(z)\rho s_t - \gamma\Phi(n)\sigma(z)^2(1-\rho^2)\right] = 0.$$

At the interior optimum, the second-order condition implies $G_n(n^*, \rho; s_t) < 0$. Hence the implicit function theorem gives

$$\frac{\partial n^*}{\partial \rho} = -\frac{G_\rho}{G_n}.$$

Differentiating G with respect to ρ gives

$$G_\rho = \beta\Phi'(n^*)\sigma(z)\left[s_t + 2\gamma\Phi(n^*)\sigma(z)\rho\right].$$

Since $\beta > 0$, $\Phi'(n^*) > 0$, $\sigma(z) > 0$, and $G_n < 0$, the sign of $\partial n^*/\partial\rho$ is the sign of

$$s_t + 2\gamma\Phi(n^*)\sigma(z)\rho,$$

which proves (20).

When z is endogenous, let

$$W(n, z; \rho, s_t) = u(m_t - n(q_t + z)) + \beta K(n, z; \rho, s_t).$$

The first-order conditions are

$$W_n(n^*, z^*; \rho, s_t) = 0, \qquad W_z(n^*, z^*; \rho, s_t) = 0.$$

Totally differentiating this system with respect to ρ yields

$$H \begin{pmatrix} \partial n^*/\partial \rho \\ \partial z^*/\partial \rho \end{pmatrix} + \begin{pmatrix} W_{n\rho} \\ W_{z\rho} \end{pmatrix} = 0,$$

where H is the Hessian of W with respect to (n, z). Direct differentiation gives

$$W_{n\rho} = \beta \Phi'(n^*)\sigma(z^*) \left[s_t + 2\gamma \Phi(n^*)\sigma(z^*)\rho \right],$$

and

$$W_{z\rho} = \beta \Phi(n^*)\sigma_z(z^*) \left[s_t + 2\gamma \Phi(n^*)\sigma(z^*)\rho \right].$$

Solving the linear system gives (21). □

A.5 Proof of Proposition 4

Proof. Wherever the worst-case belief is interior, (22) gives

$$\rho^w(n, z, s_t) = -\frac{s_t}{\gamma \Phi(n)\sigma(z)}.$$

Substituting this value into (18) gives the minimized continuation value (26),

$$\min_{\rho} K(n, z; \rho, s_t) = \Phi(n)\mu(z) - \frac{\gamma}{2}\Phi(n)^2\sigma(z)^2 - \frac{s_t^2}{2\gamma}.$$

The last term depends on the signal, but it does not depend on either choice variable, n or z.

For plans with interior worst-case beliefs, the objective equals the signal-free objective in (23) plus $-\beta s_t^2/(2\gamma)$. This term is constant with respect to the choices. It changes value but preserves the maximizer.

The signal restriction (24) is equivalent to

$$\underline{\rho} < -\frac{s_t}{\gamma \Phi(n^0)\sigma(z^0)} < \overline{\rho}.$$

The worst-case belief is interior at (n^0, z^0). Continuity keeps it interior in a neighborhood of this plan. In that neighborhood, the objective differs from the signal-free objective by a constant. The unique local maximizer therefore stays (n^0, z^0). Hence

$$n^*(s_t) = n^0, \qquad z^*(s_t) = z^0,$$

and the budget condition gives

$$c^*(s_t) = m_t - n^0(q_t + z^0).$$

The constant-shift argument extends globally only across plans whose worst-case belief is interior. For a plan whose worst case sits at a boundary $\hat{\rho} \in \{\underline{\rho}, \overline{\rho}\}$, completing the square in (18) gives

$$\min_{\rho \in [\underline{\rho}, \overline{\rho}]} K = \Phi(n)\mu(z) - \frac{\gamma}{2}\Phi(n)^2\sigma(z)^2 - \frac{s_t^2}{2\gamma} + \frac{\gamma}{2}\left[\Phi(n)\sigma(z)\hat{\rho} + \frac{s_t}{\gamma}\right]^2,$$

The last term is nonnegative and depends on the plan. Such plans can therefore change value relative to (n^0, z^0) when the signal changes. A global result requires three additional conditions. Every candidate optimizer must have an interior worst-case belief. The signal-free objective must be globally strictly concave on that set. Also, (n^0, z^0) must weakly dominate all boundary-belief and childless plans. Under these conditions, (25) is the unique global optimum.

Finally, consider the endpoints of (24). The worst-case belief at (n^0, z^0) then equals a bound $b \in \{\underline{\rho}, \overline{\rho}\}$. This plan lies at the transition between the interior and boundary cases. Define exposure $A(n, z) = \Phi(n)\sigma(z)$ and $A^0 = A(n^0, z^0)$. The endpoint signal satisfies $s_t = -\gamma b A^0$. Substitute this signal into the boundary formula. Relative to the interior formula, it adds $\frac{\gamma b^2}{2}[A(n, z) - A^0]^2$. This term is zero at (n^0, z^0). It is positive at plans with different exposure. It therefore reduces local concavity.

Let H_0 be the interior objective's Hessian at (n^0, z^0). Under the maintained negative-definiteness condition, H_0 is negative definite. After discounting by β, the added Hessian is $\beta\gamma b^2 \nabla A \nabla A'$. This matrix is positive semidefinite and has rank one. The strengthened second-order condition is $H_0 + \beta\gamma b^2 \nabla A \nabla A' \prec 0$. It preserves a strict local maximum at the endpoint. This is the condition discussed after the proposition.

In the explicit example, $\Phi(n) = n$ and z is fixed. Both Hessians are scalars. The interior Hessian is $H_0 = -\beta\gamma\sigma^2$. The added term is $\beta\gamma b^2\sigma^2$. The condition reduces to $-\beta\gamma\sigma^2(1 - b^2) < 0$. It holds because $b^2 < 1$. □

A.6 Proof of Proposition 5

Proof. Write the inner objective of (32) as

$$W(n; \rho, \alpha, s_t) = y - (q - \alpha\tau)n + \beta\left[n(\mu + \sigma\rho s_t) - \frac{\gamma}{2}n^2\sigma^2(1 - \rho^2)\right].$$

Part (i). For any (n, ρ) with $n > 0$ and $\tau > 0$,

$$\frac{\partial W}{\partial \alpha} = \tau n \geq 0,$$

Thus W is nondecreasing in α. The inner minimum is attained at $\alpha^* = \underline{\alpha}$ for every plan and ρ. If $n = 0$ or $\tau = 0$, the objective is independent of α. The same selection remains valid. The additive term $\alpha\tau n$ is independent of ρ. The joint minimization therefore separates:

$$\min_{\rho,\alpha} W(n; \rho, \alpha, s_t) = \min_{\rho \in [\underline{\rho}, \overline{\rho}]} W\big(n; \rho, \underline{\alpha}, s_t\big).$$

The resulting problem is exactly the closed-form problem of Section 3.5 with the child cost $q_e(\tau) = q - \underline{\alpha}\tau$ in place of q. All results of Sections 3.1–3.5 therefore apply with this substitution. In particular, (28) becomes (33).

Part (ii). Differentiating (33) with respect to τ gives

$$\frac{\partial n^0}{\partial \tau} = \frac{\underline{\alpha}}{\beta\gamma\sigma^2}.$$

Part (iii). Consider the uncertain program $(\tau, [\underline{\alpha}, \overline{\alpha}])$. Part (i) gives effective cost $q - \underline{\alpha}\tau$ for every plan and signal. A certain benefit of $\underline{\alpha}\tau$ with known $\alpha \equiv 1$ gives the same objective and feasible set. The argmax correspondences and policies therefore coincide. When the lowest delivery rate is positive, matching τ^* requires $\underline{\alpha}\tau = \tau^*$. Thus $\tau = \tau^*/\underline{\alpha}$.

Part (iv). By Part (i) and the definition of λ in Section 3.5,

$$\lambda(\tau) = \frac{\mu - (q - \underline{\alpha}\tau)/\beta}{\sigma}, \qquad \frac{\partial \lambda}{\partial \tau} = \frac{\underline{\alpha}}{\beta\sigma} > 0$$

when $\underline{\alpha} > 0$. By (29), width is $\lambda(\tau)(\overline{\rho} - \underline{\rho})$. To obtain the zero-delivery-bound limit, take $\underline{\alpha} \to 0$ in Part (ii). Realized spending per household is still $\alpha_0 \tau n^0 > 0$ when the realized benefit and fertility are positive. □

A.7 Dynamic consistency of the composed criterion

This subsection explains dynamic consistency after Assumption 2. A continuation plan selected at date t must remain optimal at date $t + 1$ under the information then available. A later evaluation date alone must not cause a plan change. Equation (6) combines worst-case model selection and a CARA certainty equivalent. Each part requires a separate argument.

First suppose nature selects one ρ for all dates. The worst model for the full future at date 0 can differ from the worst model for the remaining future at date 1. A new evaluation can then change the

plan. Epstein and Schneider (2003) avoid this conflict with rectangular priors. Nature can select a new one-step law at every date and history. Assumption 2 allows all measurable selections $\rho_t(h_t) \in [\underline{\rho}, \overline{\rho}]$. Any conditional law allowed at date 1 was already allowed at date 0. This gives dynamic consistency for the maxmin evaluation.

The CARA certainty equivalent $-\frac{1}{\gamma}\log \mathbb{E}[e^{-\gamma Y}]$ is nonlinear. The Epstein–Schneider theorem does not establish consistency for the combined evaluation. Following recursive robust control (Hansen and Sargent, 2008), I define preferences directly by (6). Each generation uses current utility and the same two evaluations of continuation value. There is no separate preference rule over complete future paths. The recursion defines preferences and their timing.

Appendix A.8 gives the entropy representation. In each period, the parent selects a family plan. Nature then selects ρ_t and changes conditional probability weights. The change in weights has an entropy penalty scaled by $1/\gamma$. Both evaluations concern the next generation. The next period repeats the same problem. Nature need not keep the same model or weights across generations.

A.8 Properties of the distorted measure

The distorted measure Q_t^* gives extra weight to poor future outcomes. It starts from the worst-case law $P_{\rho_t^*, z_t^*}$. Its Radon–Nikodym derivative is the factor that changes each state's weight:

$$\frac{\mathrm{d}Q_t^*}{\mathrm{d}P_{\rho_t^*, z_t^*}} = \frac{\exp\{-\gamma\Phi(n_t^*)V(x_{t+1}, s_{t+1})\}}{\mathbb{E}_{\rho_t^*, z_t^*}\left[\exp\{-\gamma\Phi(n_t^*)V(x_{t+1}, s_{t+1})\} \mid x_t, s_t\right]}.$$

First, lower continuation values receive more relative weight. For two possible next states ω_1 and ω_2,

$$\frac{(\mathrm{d}Q_t^*/\mathrm{d}P_{\rho_t^*, z_t^*})(\omega_1)}{(\mathrm{d}Q_t^*/\mathrm{d}P_{\rho_t^*, z_t^*})(\omega_2)} = \exp\left\{-\gamma\Phi(n_t^*)\big[V(\omega_1) - V(\omega_2)\big]\right\},$$

If $V(\omega_1) < V(\omega_2)$, state ω_1 receives more relative weight under Q_t^*. Larger γ increases this shift toward low-value states. As $\gamma \to 0$, the measure converges to $P_{\rho_t^*, z_t^*}$.

Second, the CARA certainty equivalent admits the entropy representation familiar from risk-sensitive and robust control,

$$-\frac{1}{\gamma}\log \mathbb{E}_{P_{\rho_t^*, z_t^*}}\left[e^{-\gamma Y_{t+1}} \mid x_t, s_t\right] = \min_{Q \ll P_{\rho_t^*, z_t^*}} \left\{\mathbb{E}_Q[Y_{t+1} \mid x_t, s_t] + \frac{1}{\gamma} D(Q \| P_{\rho_t^*, z_t^*})\right\},$$

Here $Y_{t+1} = \Phi(n_t^*)V(x_{t+1}, s_{t+1})$. Relative entropy D measures the difference between probability laws. The minimizing measure is Q_t^*. The formula balances a lower expected payoff against the penalty for changing probabilities. The resulting weights give a pessimistic evaluation. They depend on the parent's risk sensitivity and family plan. They are not fixed subjective probabilities for an expected-utility model.

A.9 Risk-neutral and ambiguity-neutral limits

The model adds ambiguity aversion and risk sensitivity to Barro–Becker. Ambiguity aversion selects the worst admissible ρ for each plan. Risk sensitivity uses a CARA certainty equivalent. Removing each feature separately gives the limits in Remark 2.

First remove risk sensitivity. Lemma 1 gives $\mathcal{C}_{\gamma,\rho,z}(Y) = \mathbb{E}_{\rho,z}[Y] - \frac{\gamma}{2}\operatorname{Var}_{\rho,z}(Y) + o(\gamma)$. As $\gamma \to 0$, the certainty equivalent approaches conditional expectation. The weight $\Phi(n_t)$ is nonnegative and chosen at time t. It can therefore move outside the expectation and minimization. The Bellman equation becomes

$$V(x_t, s_t) = \max_{c_t, n_t, z_t} \left\{ u(c_t) + \beta\Phi(n_t) \min_{\rho \in [\underline{\rho}, \overline{\rho}]} \mathbb{E}_{\rho, z_t} \left[V(x_{t+1}, s_{t+1}) \mid x_t, s_t \right] \right\}. \tag{A.1}$$

Equation (A.1) is the risk-neutral robust Barro–Becker problem. Fertility still scales the value of descendants through $\Phi(n_t)$, but the continuation value is evaluated under the worst admissible signal-informativeness parameter.

Next retain risk sensitivity and remove ambiguity. A singleton set $\underline{\rho} = \overline{\rho} = \rho$ removes the inner minimization. The parent uses one transition law and evaluates risk through the CARA certainty equivalent in (6). Finally, impose both $\gamma \to 0$ and $\underline{\rho} = \overline{\rho} = \rho$. This gives the standard Barro–Becker recursion:

$$V(x_t, s_t) = \max_{c_t, n_t, z_t} \left\{ u(c_t) + \beta\Phi(n_t) \mathbb{E} \left[V(x_{t+1}, s_{t+1}) \mid x_t, s_t \right] \right\}. \tag{A.2}$$

Thus the standard model is recovered only when the parent uses a single probability law and evaluates continuation value by an ordinary expectation.

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